THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE UNIT TEST SET 2

Chapters 11–14: **Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The cost of flooring a 18 m by 12 m hall at ₹ 45 per square metre is (in ₹): 1

(a) 261 **(b)** 9720 **(c)** 2700 **(d)** 1350

2. The net of a cylinder consists of: 1

(a) two circles and one triangle **(b)** one circle and one sector

(c) two circles and one rectangle **(d)** two rectangles and one circle

3. A dot plot shows: 1 dot at 3, 1 dot at 4, 6 dots at 5, 3 dots at 6, 9 dots at 7, 6 dots at 8. The mode of the data is: 1

(a) 3 **(b)** 7 **(c)** 4 **(d)** 5

4. The value of $8 \times 8 - 7 \times 9$ is: 1

(a) 8 **(b)** 1 **(c)** 0 **(d)** 2

5. 2 m² expressed in cm² is: 1

(a) 20000 **(b)** 200 **(c)** 200000 **(d)** 2000

6. The top view of a cone standing on its base is a: 1

(a) square **(b)** circle (with a dot at the centre)

(c) rectangle **(d)** triangle

7. The values 5, 6, 7, 10 occur 6, 3, 2, 1 times respectively. The mean is: 1

(a) 28 **(b)** 7 **(c)** 6 **(d)** 3

8. For any n, $(n+1)^2 - (n-1)^2$ equals: 1

(a) 2 **(b)** $4n$ **(c)** 4 **(d)** $2n$

9. The area of a trapezium is 70 cm². One parallel side is 12 cm and the height is 7 cm. The other parallel side (in cm) is: 1

(a) 4 **(b)** 20 **(c)** 8 **(d)** 10

10. A polyhedron has 10 vertices and 15 edges. The number of faces is: 1

(a) 5 **(b)** 27 **(c)** 7 **(d)** 23

11. The median of 18, 49, 17, 5, 24, 11, 20, 40 is: 1

(a) 20 **(b)** 18 **(c)** 19 **(d)** 23

12. Think of a number n. Add 6, multiply by 4, then subtract 24. The result is: 1

(a) $4n$ **(b)** $6n$ **(c)** $n + 6$ **(d)** $4n + 6$

13. The area of a rectangle is 352 cm² and its length is 32 cm. Its breadth (in cm) is: 1

(a) 11 **(b)** 176 **(c)** 320 **(d)** 13

14. A Koch curve starts with one segment. The number of segments at stage 2 is: 1

(a) 9 **(b)** 8 **(c)** 16 **(d)** 15

15. The mean of 6 numbers is 25. If 5 of them are 30, 22, 28, 20, 24, the remaining number is: 1

(a) 31 **(b)** 26 **(c)** 25 **(d)** 24

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. The mean of 4 numbers is 27. Three of them are 20, 31, 25. Find the fourth. 2

17. Think of a number n, add 4, multiply by 3, subtract 12, divide by 3. Show that the answer is n. 2

18. Find the area of a parallelogram whose base is 250 cm and corresponding height is 80 cm. Express it in m². 2

19. A solid has 9 faces and 21 edges. Use Euler's formula to find its number of vertices. 2

20. Each of the data values 25, 30, 27, 15, 23 is increased by 4. Find the old and the new mean. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. The parallel sides of a trapezium are 17 cm and 9 cm and its area is 130 cm². Find the distance between the parallel sides. 3

22. A Sierpinski triangle starts with a triangle of area 64 cm². Find the shaded area after stage 2 and the area removed. 3

23. The mean of 5 observations is 24. One observation, 12, is removed. Find the new mean. 3

24. A 3×3 block of calendar dates has middle date 11. Write the nine dates and show that their sum is 9 times the middle date. 3

OR

The diagonals of a rhombus are 12 cm and 16 cm. Find its area and its side. 3

25. A cube of side 4 cm is painted on all faces and cut into 1 cm cubes. Find the number of small cubes with
(i) exactly two faces painted
(ii) exactly one face painted. 3

OR

The runs scored by a batter in 6 innings are 39, 16, 28, 33, 82, 90. Find the mean, the median and the range. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A cube of side 9 cm is painted and cut into 1 cm cubes. Find the number of cubes with three, two, one and no painted faces. Verify that these add up to 729. 5

OR

The runs of two players in 5 matches are: A: 80, 18, 34, 43, 30; B: 39, 49, 41, 20, 36. Find the mean of each and say who is more consistent (smaller range). 5

27. Think of a number n. Double it, add 10, halve the result and subtract n. Show algebraically that the final answer is always 5. Verify for $n = 7$. 5

OR

A rectangular room 12 m by 9 m is to be carpeted with a carpet 2 m wide costing ₹ 80 per metre. Find the length of carpet needed and the total cost. 5

END OF QUESTION PAPER **Total: 50 Marks**

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